



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DDL Commands – CREATE, ALTER, DROP

EXPERIMENT – 1

Ex. No.: 1

Date:

AIM:

To Create, Alter, Drop, Rename and Truncate tables using Data Definition Language (DDL) statements.

DESCRIPTION:

Data Definition Language (DDL) statements are used to define the database structure or schema.

DDL Commands:

- **CREATE** – Creates database objects such as tables.
- **ALTER** – Modifies the structure of an existing table.
- **DROP** – Deletes a table from the database.
- **TRUNCATE** – Removes all records from a table.
- **RENAME** – Renames a table.

SYNTAX

CREATE TABLE

CREATE TABLE table_name

(

column_name1 datatype,

column_name2 datatype,

column_name3 datatype,

...

);

ALTER TABLE

Add a Column

ALTER TABLE table_name

ADD column_name datatype;

Drop a Column

ALTER TABLE table_name

DROP COLUMN column_name;

DROP TABLE

DROP TABLE table_name;

TRUNCATE TABLE

TRUNCATE TABLE table_name;

RENAME TABLE

RENAME TABLE old_table_name TO new_table_name;

QUESTIONS

1) Create a table named HOSTEL_ROOMS with the following structure.

S.No	Column Name	Description	Data Type
1	RoomID	Room Identifier	INT
2	RoomNumber	Room Number	VARCHAR(10)
3	BlockName	Hostel Block Name	VARCHAR(20)
4	FloorNo	Floor Number	INT
5	RoomType	Room Type	VARCHAR(20)
6	Capacity	Maximum Capacity	INT
7	OccupiedBeds	Occupied Beds	INT
8	RentPerMonth	Monthly Rent	DECIMAL(8,2)
9	RoomStatus	Room Status	VARCHAR(15)
10	WardenName	Warden Name	VARCHAR(50)

Answer

CREATE DATABASE IF NOT EXISTS HostelDB;

USE HostelDB;

CREATE TABLE Hostel_Rooms

(

RoomID INT PRIMARY KEY,

RoomNumber VARCHAR(10),

BlockName VARCHAR(20),

FloorNo INT,

RoomType VARCHAR(20),

Capacity INT,

OccupiedBeds INT,

RentPerMonth DECIMAL(8,2),

RoomStatus VARCHAR(15),

WardenName VARCHAR(50)

);

The screenshot shows a SQL IDE interface with a script editor and a result grid. The script editor contains the following SQL code:

```
10      OccupiedBeds INT,  
11      RentPerMonth DECIMAL(8,2),  
12      RoomStatus VARCHAR(15),  
13      WardenName VARCHAR(50)  
14  );  
15  • SHOW COLUMNS FROM Hostel_Room;  
16  • ALTER TABLE Hostel_Room  
17    ADD RoomPhone BIGINT;  
18  • DESC Hostel_Room;
```

The result grid displays the table structure for Hostel_Room:

	Field	Type	Null	Key	Default	Extra
▶	RoomID	int	NO	PRI		
	RoomNumber	varchar(10)	YES			
	BlockName	varchar(20)	YES			
	FloorNo	int	YES			
	RoomType	varchar(20)	YES			
	Capacity	int	YES			
	OccupiedBeds	int	YES			
	RentPerMonth	decimal(8,2)	YES			
	RoomStatus	varchar(15)	YES			
	WardenName	varchar(50)	YES			
	RoomPhone	bigint	YES			

The output pane shows the execution of the ALTER and DESC statements:

#	Time	Action	Message
22	11:54:23	ALTER TABLE Hostel_Room ADD RoomPhone BIGINT	0 row(s) affected Records: 0 Dup
23	11:54:31	DESC Hostel_Room	11 row(s) returned

2) Create a table named HOSTEL_STUDENT with the following structure.

S.No	Column Name	Description	Data Type
1	StudentID	Student Identifier	INT
2	StudentName	Student Name	VARCHAR(30)
3	Gender	Gender	CHAR(1)
4	DOB	Date of Birth	DATE
5	MobileNo	Mobile Number	BIGINT
6	RoomID	Room Identifier	INT

Answer

CREATE TABLE Hostel_Student

(

StudentID INT PRIMARY KEY,

StudentName VARCHAR(30),

Gender CHAR(1),

DOB DATE,

MobileNo BIGINT,

RoomID INT

)

The screenshot shows the SQL File 5* editor with the following SQL code:

```

42     BlockName VARCHAR(20),
43     TotalRooms INT,
44     FloorCount INT,
45     WardenID VARCHAR(5)
46 );
47
48 • ALTER TABLE Hostel_Student
49   ADD Email VARCHAR(50);
50 • DESC Hostel_Student;
  
```

Below the editor, the 'Result Grid' is visible, showing the structure of the 'Hostel_Student' table:

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
StudentName	varchar(30)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
RoomID	int	YES		NULL	
Email	varchar(50)	YES		NULL	

3) Create a table named **HOSTEL_WARDEN** with the following structure.

S.No	Column Name	Description	Data Type
1	WardenID	Warden Identifier	VARCHAR(5)
2	WardenName	Warden Name	VARCHAR(30)
3	Gender	Gender	CHAR(1)
4	MobileNo	Mobile Number	BIGINT
5	Experience	Years of Experience	INT

Answer

CREATE TABLE Hostel_Warden

(
WardenID VARCHAR(5) PRIMARY KEY,

WardenName VARCHAR(30),

Gender CHAR(1),

MobileNo BIGINT,

Experience INT

);

SQL File 3* SQL File 3* SQL File 4* x SQL File 5*

Limit to 1000 rows

```

42 BlockName VARCHAR(20),
43 Execute the selected portion of the script or everything, if there is no selection
44 FloorCount INT,
45 WardenID VARCHAR(5)
46 );
47
48 • ALTER TABLE Hostel_Student
49 ADD Email VARCHAR(50);
50 • DESC Hostel_Warden;

```

Result Grid Filter Rows: Export: Wrap Cell Content: A

Field	Type	Null	Key	Default	Extra
WardenID	varchar(5)	NO	PRI	NULL	
WardenName	varchar(30)	YES		NULL	
Gender	char(1)	YES		NULL	
MobileNo	bigint	YES		NULL	
Experience	int	YES		NULL	

Result 2 x Read Only

4) Create a table named **HOSTEL_BLOCK** with the following structure.

S.No	Column Name	Description	Data Type
1	BlockID	Block Identifier	VARCHAR(5)
2	BlockName	Block Name	VARCHAR(30)
3	TotalRooms	Total Number of Rooms	INT
4	FloorCount	Number of Floors	INT
5	WardenID	Warden Identifier	VARCHAR(5)

Answer

CREATE TABLE Hostel_Block

(

BlockID VARCHAR(5) PRIMARY KEY,

BlockName VARCHAR(30),

TotalRooms INT,

FloorCount INT,

WardenID VARCHAR(5)

);

The screenshot shows a SQL IDE with four tabs: SQL File 3*, SQL File 3*, SQL File 4* (active), and SQL File 5*. The active tab contains the following SQL code:

```
42     BlockName VARCHAR(20),
43     TotalRooms INT,
44     FloorCount INT,
45     WardenID VARCHAR(5)
46 );
47
48 • ALTER TABLE Hostel_Student
49   ADD Email VARCHAR(50);
50 • DESC Hostel_Block;
```

Below the code editor, the 'Result Grid' is displayed, showing the structure of the 'Hostel_Block' table:

	Field	Type	Null	Key	Default	Extra
►	BlockID	varchar(5)	NO	PRI	NULL	
	BlockName	varchar(20)	YES		NULL	
	TotalRooms	int	YES		NULL	
	FloorCount	int	YES		NULL	
	WardenID	varchar(5)	YES		NULL	

On the right side of the IDE, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, there is a 'Result 3' tab and a 'Read Only' indicator.

5) Modify the table HOSTEL_STUDENT by adding a column named Email of datatype VARCHAR(50).

Answer

ALTER TABLE Hostel_Student

ADD Email VARCHAR(50);

Additional DDL Commands

Rename Table

RENAME TABLE Hostel_Block TO Hostel_Blocks;

Truncate Table

TRUNCATE TABLE Hostel_Blocks;

Drop Table

DROP TABLE Hostel_Blocks;

The screenshot shows a MySQL IDE interface with four tabs: 'SQL File 3*', 'SQL File 3*', 'SQL File 4*' (active), and 'SQL File 5*'. The active tab contains the following SQL code:

```
42      BlockName VARCHAR(20),
43      TotalRooms INT,
44      FloorCount INT,
45      WardenID VARCHAR(5)
46  );
47
48  • ALTER TABLE Hostel_Student
49    ADD Email VARCHAR(50);
50  • DESC Hostel_Student;
```

Below the code editor is a 'Result Grid' section. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The table structure is displayed as follows:

	Field	Type	Null	Key	Default	Extra
►	StudentID	int	NO	PRI	NULL	
	StudentName	varchar(30)	YES		NULL	
	Gender	char(1)	YES		NULL	
	DOB	date	YES		NULL	
	MobileNo	bigint	YES		NULL	
	RoomID	int	YES		NULL	
	Email	varchar(50)	YES		NULL	

On the right side of the interface, there is a vertical toolbar with icons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom right, there is a 'Read Only' indicator.

RESULT

Thus, the Data Definition Language (DDL) commands CREATE, ALTER, RENAME, TRUNCATE, and DROP were successfully executed on the Hostel Management System tables in MySQL.